

```
1 arr=[10,25,30,45,50]
2 key=30
3 found=False
4 for i in range(len(arr)):
5     if arr[i] == key:
6         print("key found at index",i)
7         found=True
8         break
9 if not found:
10     print("key not found")
```

key found at index 2

=== Code Execution Success ===

main.py



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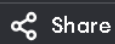
Output

```
1 def binary_search(arr, low, high, key):
2     if low > high:
3         return -1
4     mid = (low + high) // 2
5     if arr[mid] == key:
6         return mid
7     elif key < arr[mid]:
8         return binary_search(arr, low, mid - 1, key)
9     else:
10        return binary_search(arr, mid + 1, high, key)
11 arr = [5, 10, 15, 20, 25]
12 key = 20
13 index = binary_search(arr, 0, len(arr) - 1, key)
14 if index == -1:
15     print("Key not found")
16 else:
17     print("Key found at index", index)
```

Key found at index 3

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Output

```
1 n=5
2 fact=1
3 for i in range(1,n+1):
4     fact*=i
5 print("Iterative:",fact)
6 def factorial(n):
7     if n==0 or n==1:
8         return 1
9     return n*factorial(n-1)
10 print("Recursive:",factorial(n))
```

Iterative: 120

Recursive: 120

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```
1 n=6
2 a,b=0,1
3 for i in range(n):
4     print(a,end=" ")
5     a,b=b,a+b
```

Output

0 1 1 2 3 5

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main.py



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Output

```
1 A=[[1,2],[3,4]]
2 B=[[5,6],[7,8]]
3 result=[[0,0],[0,0]]
4 for i in range(2):
5     for j in range(2):
6         for k in range(2):
7             result[i][j]+=A[i][k]*B[k][j]
8 print(result)
```

[[19, 22], [43, 50]]

=== Code Execution Successful ===